Stage 3: Model Fitting, Network Size, Identifiability, and Robust Vaccination Thresholds

Q1. Why does the fit need a “network size” at all?

Section 4.2 considers a South Carolina community of 15,000 people with 17% two-dose coverage. At R0 = 15, the unmitigated model predicts approximately 12,600–13,200 cases, versus 990 in the observed series. Using 13,000 as a representative prediction:

$$13{,}000 - 990 = 12{,}010 \text{ excess predicted cases; } 13{,}000 / 990 = 13.13 \text{ times the observed total.}$$

For a homogeneous, closed, initially susceptible epidemic without intervention, the nonzero final-size solution satisfies:

$$z = 1 - \exp(-Rz); \text{ with } R = 12.5: z = 1 - \exp(-12.5z).$$

Trial z	1 – exp(–12.5z)	RHS – z
0.900000	0.999986993	+0.099986993
0.990000	0.999995777	+0.009995777
0.999000	0.999996226	+0.000996226
0.999990	0.999996273	+0.000006273
0.999999	0.999996273	-0.000002727

The nonzero root is approximately z = 0.99999627, or 99.99963% of the initially susceptible population. The zero solution is mathematically possible but is not the epidemic final-size branch. This relation is an illustrative homogeneous approximation, not an exact final-size equation for the heterogeneous, leaky-vaccine model.

In a closed, unmitigated SEIR epidemic, susceptibles are depleted as infections occur. In the homogeneous approximation, $R_{eff}(t) = R\,S(t)/S(0)$, so transmission becomes subcritical once $S(t)/S(0) < 1/12.5 = 0.08$. The epidemic does not simply fade while an abundant, well-connected susceptible pool remains.

The fit therefore estimates an effective accessible community network, N2eff. In South Carolina, N2eff = 1,277 (95% profile interval 1,060–1,940), which is 1,277/15,000 × 100 = 8.51% of the nominal community. This is an effective transmission-network size, not a census measurement.

A competing explanation is the public-health response: the draft reports 874 students asked to quarantine and 3,788 additional MMR doses administered in Spartanburg County. Restricted network reach and interventions can both shorten an outbreak, so they are confounded. A constant-transmission fit attributes the early ending to N2eff even if interventions contributed.

Q2. Work through Proposition 3(i) yourself.

From equation (5), K = [[a,b],[c,d]], with:

$$a = R0\ s1[\theta + (1-\theta)p1];\ b = R0\ s1(1-\theta)p1;$$

$$c = R0\ s2(1-\theta)p2;\ d = R0\ s2[\theta + (1-\theta)p2].$$

As $p2 \to 0$ and $p1 \to 1$, the entries approach:

$a \to R0\ s1$; $b \to R0\ s1(1-\theta)$; $c \to 0$; $d \to R0\ \theta\ s2$.

The limiting matrix is upper triangular. Since $c = 0$, $bc = 0$ and $\sqrt{((a-d)^2 + 4bc)} = \sqrt{((a-d)^2)} = |a-d|$. Therefore:

$Rv \to (a+d+|a-d|)/2 = \max(a,d)$.

Proof by cases: if $a \geq d$, then $|a-d| = a-d$ and the expression equals $(a+d+a-d)/2 = a$. If $d > a$, then $|a-d| = d-a$ and it equals $(a+d+d-a)/2 = d$.

Check 1: $a = 2$, $d = 5 \to (2+5+3)/2 = 5 = \max(2,5)$.

Check 2: $a = 5$, $d = 2 \to (5+2+3)/2 = 5 = \max(5,2)$.

Hence $Rv \to \max(R0\ s1, R0\ \theta\ s2)$.

In words, as the community becomes a negligible share of the surrounding population, its infections have little effect on general-population totals, although the community may still sustain substantial internal transmission.

Q3. Why is h* the “robust” quantity?

The West Texas coverage sensitivity results from the supplied script are:

Assumed coverage	Fitted Rv	Critical c2*	Susceptible share h*
10%	3.00	74.0%	68.7%
30%	3.00	80.0%	68.4%
50%	2.99	86.0%	67.8%

The critical coverage changes by 86.0 – 74.0 = 12.0 percentage points, while h* changes by only 68.7 – 67.8 = 0.9 percentage points. Rv remains approximately 3.00.

Proposition 3(iii) defines $h^* = 1 - s2^*/s2$. For a small community with community-dominated transmission, $s2^* \approx 1/(R0\ \theta)$ and $Rv \approx R0\ \theta\ s2$. Substitution yields:

$h^* \approx 1 - 1/(R0\ \theta\ s2) \approx 1 - 1/Rv$.

At $Rv = 3.00$: $h^* \approx 1 - 1/3 = 2/3 = 66.67\%$.

The exact fitted West Texas value is 68.4% (95% profile interval 61.3–74.3%), which is 1.73 percentage points higher than the small-community approximation. The difference reflects finite community size and nonzero general-population susceptibility.

With two-dose efficacy $\varepsilon 2 = 0.97$, $s2 = 1 - 0.97c2$. Under assumed coverage of 10%, 30%, and 50%, the susceptible fractions are 0.903, 0.709, and 0.515, respectively. To retain $Rv \approx 3.00$, the approximation $R0\theta \approx Rv/s2$ requires:

Coverage	s2 = 1 – 0.97c2	Approximate R0θ
10%	0.903	3.00/0.903 = 3.32
30%	0.709	3.00/0.709 = 4.23
50%	0.515	3.00/0.515 = 5.83

These are illustrative values implied by the approximation, not separately reported refits. Higher assumed coverage means fewer susceptibles, so fitted $R0\theta$ must increase to reproduce the same Rv. The case curves constrain effective transmission better than they identify baseline community coverage.

The robust h* answers what fraction of those currently susceptible must become effectively immune. It is more stable than the absolute two-dose coverage threshold when initial coverage is uncertain. It is not automatically the fraction needing a vaccine dose, because actual campaign dose requirements depend on vaccine efficacy and who is already protected.